

Underneath a

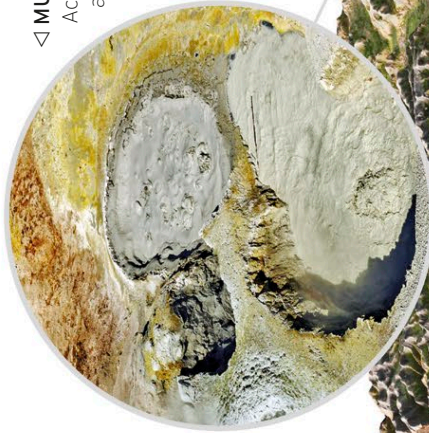
SUPERVOLCANO

There are a handful of volcanic sites in the world known as supervolcanoes: sites that have seen at least one eruption rating an 8 on a scale called the Volcanic Explosivity Index. These are the largest eruptions that geologists know about; intervals between them tend to be measured in tens or hundreds of thousands of years. These sites are also capable of future eruptions that could radically alter landscapes and

severely impact the world's climate. All supervolcanoes are calderas that sit above large, active magma chambers. Yellowstone National Park lies over a supervolcano that last erupted about 640,000 years ago. There are two large, interconnected magma chambers under Yellowstone caldera. The heat from the upper chamber drives geothermal features: hot springs, geysers, fumaroles, and boiling mud pools.

◁ MUDPOT

Acid hot springs, such as Sulfur Caldron, dissolve nearby ground into a pool of boiling, sulfurous-smelling mud, called a mudpot.



center of caldera is about 6 miles (10 km) east of Old Faithful geyser

rim of Yellowstone caldera

Teton Range

Old Faithful geyser

Grand Prismatic spring

△ COLORFUL POOLS

The colors in hot spring pools and around their edges come from pigments produced by heat-loving microorganisms or from minerals.



the Absaroka Range is one of several mountain ranges of the Rocky Mountains that surround the caldera

▷ HOT SPRING TERRACES

Hot springs can create terraces of travertine (a form of limestone) as seen at Mammoth Hot Springs, just outside the caldera boundary.

Yellowstone Lake



Upper crustal magma reservoir

This contains hot, mostly spongelike solid rock and about 9 percent magma at a temperature of around 1,400 °F (760 °C).

